Industrial Silicon

Fundamentals, Price Forecasting and Producer Risk

A Focus on the Chinese Market

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Contents

Executive Summary **3**

1 Introduction **4**

1.1 What is industrial silicon? 4

1.2 Why a futures market? 4

1.3 Questions and structure of this report 5

2 Production and Supply **6**

2.1 How industrial silicon is produced 6

2.2 Global and Chinese capacity 6

2.3 Regional structure 7

2.4 Producers and industry structure 8

3 Downstream Demand **9**

3.1 Overview 9

3.2 Polysilicon: the growth engine 10

3.3 Organosilicon 10

3.4 Aluminium–silicon alloys 11

4 Supply–Demand Balance **12**

4.1 Method 12

4.2 Results 12

5 Production Costs **14**

5.1 The cost model 14

5.2 Regional and company cost structures 14

5.3 Controlling electricity costs 15

5.4 Electricity sensitivity 16

6 Cost Curve, Margins and Stress Tests **17**

6.1 The industry cost curve 17

6.2 Margins at reference prices 17

6.3 Stress tests 18

7 Price Forecasting with ARIMA **20**
7.1 Data, and a hidden contract switch . . . 20
7.2 How ARIMA works . . . 21
7.3 Stationarity and model identification . . . 21
7.4 Out-of-sample evaluation . . . 22
7.5 Interpretation . . . 23

8 Volatility Forecasting with GARCH **24**
8.1 Why model volatility? . . . 24
8.2 The models . . . 24
8.3 In-sample fit . . . 25
8.4 Out-of-sample evaluation . . . 26
8.5 Interpretation . . . 27
8.6 Application: how likely are producers to lose money next month? . . . 27

9 Conclusions **29**
9.1 Summary of models . . . 29
9.2 Implications . . . 29
9.3 Limitations . . . 30
9.4 Future work . . . 30

A Data Sources and Glossary **31**
A.1 Data . . . 31
A.2 Glossary of market terms . . . 32

Executive Summary

Industrial silicon (metallurgical-grade silicon) is the feedstock for solar-grade polysilicon, silicone chemicals and aluminium alloys. China holds over 80% of world capacity, and futures have traded on the Guangzhou Futures Exchange since December 2022. This report combines a bottom-up analysis of supply, demand and production costs with a statistical study of futures prices and their volatility.

Fundamentals.

- **Supply is concentrated and power-hungry.** Electricity is 37–50% of production cost, which makes captive power plants and seasonal hydropower the main sources of cost advantage. Full production cost ranges from about 8,300 CNY/t at the lowest-cost producer (Hoshine, Xinjiang) to 12,600 CNY/t in Southwest China's dry season.
- **The market was persistently oversupplied.** Estimated supply exceeded demand in 33 of the 35 months from mid-2022 to mid-2025, and spot prices fell by more than half.
- **By mid-2025, almost every producer was losing money.** On 4 June 2025, Xinjiang 553# spot reached 7,600 CNY/t, below the cost of even the lowest-cost producer. Stress tests show that a 10% price fall, a 0.05 CNY/kWh rise in power prices or a 20% rise in raw-material prices each cost a producer 550–1,000 CNY/t of margin.

Statistical analysis.

- **The direction of daily price moves is not predictable.** ARIMA models do not beat a random walk ("tomorrow's price equals today's") by a statistically significant margin, whether applied to prices or to percentage returns, and open interest and trading volume carry no next-day signal.
- **The size of price moves is predictable.** Volatility clusters, and a GARCH(1,1) model with fat-tailed Student-t errors gives the best volatility forecasts and the only 95% risk intervals that pass standard backtests.
- **A data problem had hidden this.** The vendor's continuous futures series contains a false +30% one-day jump caused by a contract switch on 18 October 2024. Correcting it reveals the volatility clustering that the original analysis missed.

Bringing the two together. Simulating one month of prices with the fitted GARCH model gives the probability that each type of producer is loss-making a month ahead: about one in three for Hoshine, three in four for an average Xinjiang plant and nearly certain for Southwest plants in the dry season. These figures assume prices start from mid-July 2025 levels, just after a policy-driven rally that no time-series model anticipated.

Chapter 1

Introduction

1.1 What is industrial silicon?

Industrial silicon, also called metallurgical-grade silicon (MG-Si), is elemental silicon refined to a purity of roughly 98.5–99.9%. It is made by reducing high-purity quartz with carbon in submerged arc furnaces at temperatures above 1,700 °C. Grades are named by their maximum content of three impurities, iron, aluminium and calcium: grade 553# allows at most 0.5% Fe, 0.5% Al and 0.3% Ca, for example.

Table 1.1: Grades of industrial silicon

Grade	Purity (% Si)	Impurity limits (Fe / Al / Ca)	Typical application
553#	98.5–98.7	0.5% / 0.5% / 0.3%	Aluminium alloys, steel deoxidiser
441#	98.9–99.1	0.4% / 0.4% / 0.1%	Automotive castings, rubber, magnesium alloys
421#	99.2–99.4	0.4% / 0.2% / 0.1%	Polysilicon and organosilicon feedstock
3303#	$\geq$ 99.3	0.3% / 0.3% / 0.03%	Chemical-grade silicon
2202#	$\geq$ 99.4	0.2% / 0.2% / 0.02%	High-purity solar-grade feedstock
97#	97.0–98.5	not standardised	Low-end metallurgy

Three downstream industries absorb almost all output: **polysilicon** for solar cells and semiconductors, **organosilicon** (silicone) chemicals used in sealants, rubbers and coatings, and **aluminium–silicon alloys** used in vehicles, construction and solar module frames.

1.2 Why a futures market?

Industrial silicon is energy-intensive and capital-intensive, and its price is volatile: demand swings with solar installations, costs swing with electricity prices, and policy can change supply quickly. China is both the dominant producer and the dominant consumer, holding over 80% of world capacity in 2024. Industrial silicon futures were listed on the **Guangzhou Futures Exchange (GFEX)** on 22 December 2022, its first product. Futures let producers and users hedge price risk, let investors trade views on supply and demand, and provide a transparent reference price for the industry.

1.3 Questions and structure of this report

The report asks two kinds of questions:

1. **Fundamental:** how is industrial silicon produced, what does it cost in different regions, how do supply and demand balance, and how exposed are producers' margins to changes in electricity, raw-material and product prices? (Chapters 2–6)
2. **Statistical:** can past prices forecast future prices, and can past volatility forecast future volatility? (Chapters 7–8)

Chapter 9 brings the two together and lists open questions. Data sources and a glossary of Chinese market terms are in Appendix A.

Chapter 2

Production and Supply

2.1 How industrial silicon is produced

Quartz (silica, SiO_2) is reduced by carbon in a submerged arc furnace:

$$SiO_2 + 2\,C \longrightarrow Si + 2\,CO\uparrow \tag{2.1}$$

The carbon comes from a blend of reductants: washed coal or coke, petroleum coke and wood chips. From molar masses (SiO_2 60.1, C 12.0, Si 28.1 g/mol), one tonne of silicon in theory needs 2.14 t of silica and 0.86 t of carbon. Heat losses, side reactions and incomplete reduction push actual consumption much higher:

Table 2.1: Theoretical and actual input consumption per tonne of silicon

Input	Theoretical	Typical actual
Silica	2.14 t	2.6–2.8 t
Carbon reductants	0.86 t	1.6–2.0 t
Graphite electrode	–	0.08–0.10 t
Electricity	–	11,000–12,500 kWh

Two features of the process shape the industry's economics. First, **electricity is the single largest cost**: at 11,000–12,500 kWh per tonne, a 0.05 CNY/kWh difference in power price changes total cost by about 600 CNY/t. Second, **the choice of reductant determines the grade**. All-coal reductants leave more iron, aluminium and calcium in the product, yielding lower grades such as 553#; wood chips and petroleum coke give purer silicon suitable for chemical and solar uses.

2.2 Global and Chinese capacity

Global industrial silicon capacity reached 8.81 million tonnes in 2024, up 3.2% year on year (China Photovoltaic Industry Association). China accounted for 82.6% of it, adding 272,000 t of capacity in 2024, mostly projects deferred from 2023 and vertically integrated developments. Capacity outside China barely grew. Other producers include Brazil, the United States, Norway, France, Russia and Germany.

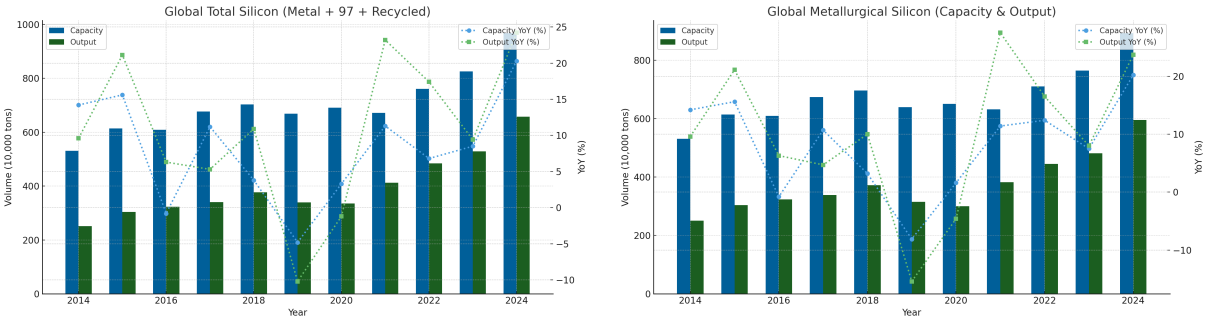


Figure 2.1: Global silicon capacity and output: all silicon including 97# and recycled (left), and metallurgical silicon only (right), with year-on-year growth.

At the end of 2024, China's capacity stood at **7.14 million tonnes** and output reached **4.9 million tonnes**, a utilisation rate of about 69%. Growth came from commissioning of earlier projects, upgrades of existing plants, and demand for exchange-deliverable grades created by the futures market.

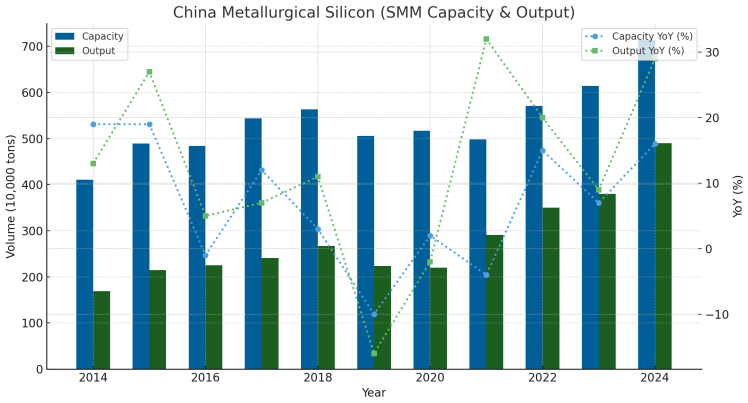


Figure 2.2: China's industrial silicon capacity and output, 2014–2024 (SMM).

2.3 Regional structure

Because electricity dominates costs, capacity clusters where power is cheap. There are two main clusters, which differ in energy source, seasonality and product quality.

- **Xinjiang** (about 38–40% of national capacity) runs on stable, low-cost coal-fired power at 0.26–0.33 CNY/kWh, much of it from producers' own power plants. Its all-coal reductants mostly yield 553# metallurgical grade for aluminium and steel users.
- **Yunnan** (22–25%) depends on hydropower. Plants run hard in the wet season (May–October), when power can fall below 0.2 CNY/kWh, and cut output in the dry season. Wood chips and petroleum coke give higher grades (421#, 3303#) for chemical and solar users.
- **Sichuan** (12–14%) is similar to Yunnan in seasonality and product mix, and several producers are integrated into downstream organosilicon production.

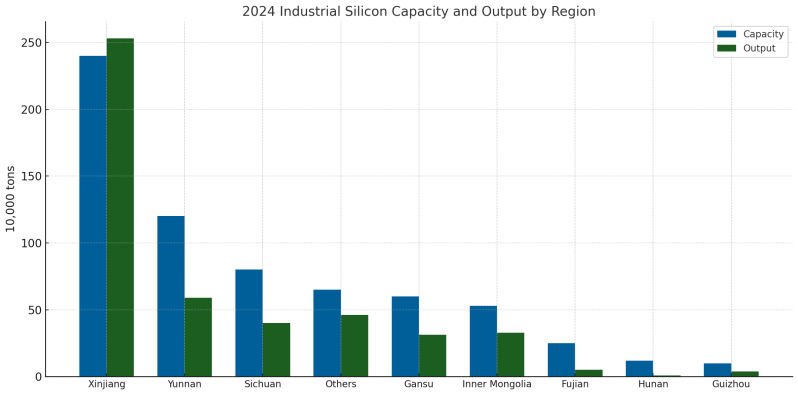


Figure 2.3: Industrial silicon capacity and output by region, 2024 (10,000 t). Capacity and output come from different sources; Xinjiang's reported output slightly exceeds its reported capacity.

2.4 Producers and industry structure

Nominal capacity is fragmented, but actual output is concentrated. In mid-2024 the five largest producers held about 35% of designed capacity but produced an estimated 60–65% of output. Large producers run integrated, energy-secure plants at high utilisation, while smaller ones are more exposed to price falls and tighter regulation. Over 800,000–1,000,000 t of capacity was under construction or announced in 2024, mainly in Inner Mongolia, Gansu and Ningxia, but with prices near the cost line, many new entrants find it uneconomic to start production.

Table 2.2: Major industrial silicon producers in China (2025)

Company	Main regions	Monthly output (kt)
Hoshine	Xinjiang, Yunnan	181
East Hope	Gansu, Qinghai	66
GCL	Inner Mongolia, Xinjiang	36
Tongwei	Sichuan	28
TBEA	Xinjiang, Inner Mongolia	14

Three trends are reshaping the industry:

- **Vertical integration.** Hoshine is integrated into organosilicon and Tongwei into polysilicon and wafers. Integration secures feedstock, lowers costs and is favoured in project approvals.
- **Tighter environmental standards** on exhaust treatment, furnace efficiency and carbon intensity raise the bar for small producers.
- **Scale and relocation to the Northwest.** New plants use furnaces of 33,000 kVA or larger and cluster in Xinjiang, Gansu, Qinghai and Inner Mongolia, where power, land and policy support are available. New approvals in power-constrained Yunnan and Sichuan have slowed.

Chapter 3

Downstream Demand

3.1 Overview

The mix of demand has shifted sharply towards solar. The polysilicon industry's share of industrial silicon consumption rose from about 19% in 2019 to 49% in 2024. Organosilicon has held a steady 30–40%, while aluminium alloys fell from over 40% to about 16%.

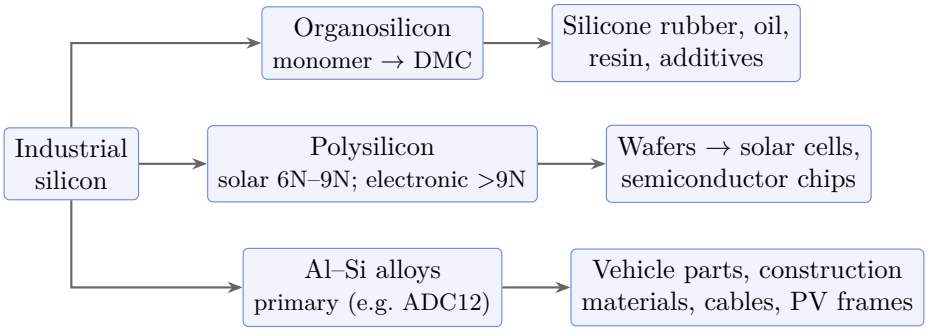


Figure 3.1: The three downstream uses of industrial silicon.

Because each downstream product uses a predictable amount of silicon per tonne, demand can be estimated from downstream output. The conversion factors are derived in the sections below.

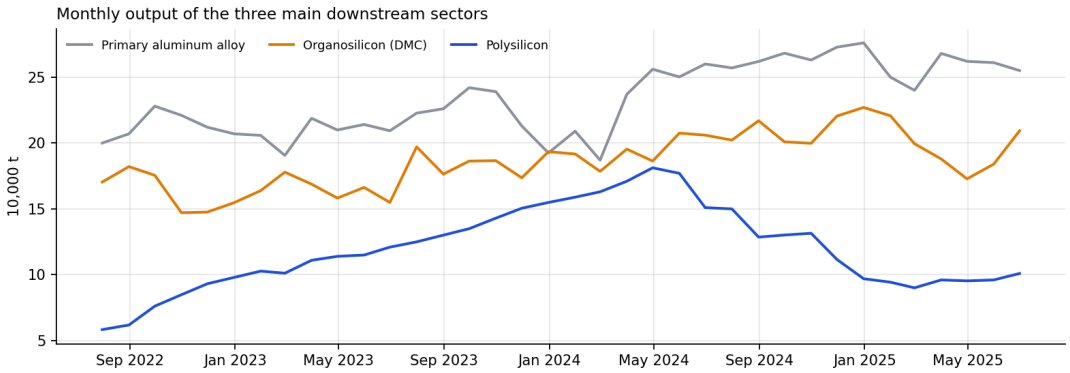


Figure 3.2: Monthly output of the three main downstream sectors (SMM).

3.2 Polysilicon: the growth engine

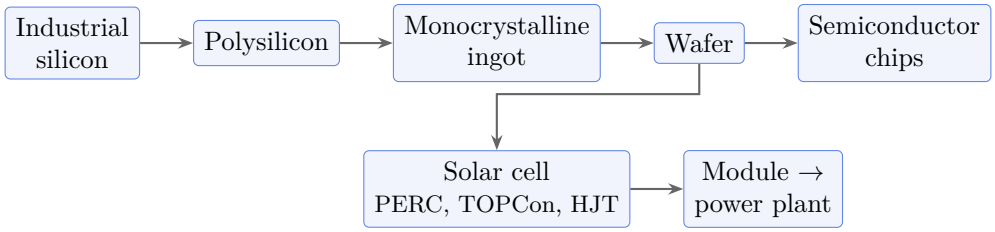


Figure 3.3: The polysilicon value chain.

Polysilicon is purified silicon used in over 95% of the world's solar cells. Solar grade has a purity of 6N to 9N (99.9999% to 99.9999999%), electronic grade above 9N. Two processes dominate: the modified Siemens process (chunk polysilicon, the mainstream) and the fluidised bed reactor (granular silicon, which uses less energy). At the end of 2024 global polysilicon capacity was 3.25 million tonnes and output 1.96 million tonnes, of which chunk silicon made up 83.4%, granular 14.6% and electronic grade 2.0%. The top five producers, all Chinese, made 70.3% of world output. N-type polysilicon, which allows more efficient cells but needs higher purity, passed 60% of the market by May 2024.

Demand for industrial silicon from this sector can be estimated in two ways:

- **Top-down, from solar installations.** Multiply new capacity (in watts) by the silicon needed per watt (Table 3.1).
- **From polysilicon output**, which is more direct. Each tonne of polysilicon uses 1.04–1.20 t of industrial silicon after processing losses; this report uses **1.08**.

Table 3.1: Polysilicon needed per watt of solar cells, and the effect of a 1,000 CNY/t rise in the silicon price on module cost

Cell type	Wafer thickness (mm)	Efficiency	g/W	Added cost (CNY/W)
M10-182 PERC	0.150	23.5%	2.01	0.0021
G12-210 PERC	0.150	23.4%	2.02	0.0021
183R	0.125	25.5%	1.62	0.0017
G12-210 TOPCon	0.125	25.6%	1.62	0.0017
HJT	0.120	25.5%	1.58	0.0016
210R	0.125	25.9%	1.60	0.0017

Silicon needed per watt is wafer mass (area × (thickness + 0.05 mm sawing loss) × density 2.33 g/cm^3) divided by the cell's power (area × efficiency × 1,000 W/m^2), adjusted for yields and losses in ingot pulling. Thinner wafers and more efficient cells have cut it from about 2.0 to 1.6 g/W. The last column shows why **solar demand hardly reacts to the silicon price**: a 1,000 CNY/t rise adds about 0.002 CNY per watt, a negligible share of module cost.

3.3 Organosilicon

Organosilicon chemicals start with the Müller–Rochow process, in which silicon reacts with chloromethane over a copper catalyst to make dimethyldichlorosilane, the main monomer:

$$2\,\mathrm{CH_3Cl} + \mathrm{Si} \longrightarrow \mathrm{Si(CH_3)_2Cl_2} \tag{3.1}$$

Hydrolysis and polymerisation of the monomer give cyclic siloxanes, collectively called DMC, $[(CH_3)_2SiO]_n$ with $n = 3$–6, the feedstock for silicone rubbers, oils and resins. End uses are fragmented (construction sealants, personal care, electronics), so demand is tracked through DMC output. One tonne of DMC needs about two tonnes of monomer, and each tonne of monomer about 0.26 t of silicon, giving **0.52 t of silicon per tonne of DMC**. Efficient plants use 0.47–0.49 t; older processes up to 0.60 t.

3.4 Aluminium–silicon alloys

Cast alloys (ADC12, A356, A380) contain 7–13% silicon to improve fluidity in casting and are used mainly for engine and vehicle parts. Wrought alloys (4xxx, 6xxx series) contain 0.2–1.7% and are used in window frames, curtain walls and solar module frames. Recycled alloys largely retain their silicon, so new silicon demand comes from **primary** alloys, at a weighted average of about **0.22 t of silicon per tonne**. Electric vehicles and solar frames are replacing demand lost from internal-combustion engines.

Table 3.2: Downstream output and implied industrial silicon demand, first half of 2025 (10,000 t)

	Polysilicon		DMC		Primary Al alloy		Implied
Month	Output	YoY	Output	YoY	Output	YoY	Si demand
Jan	9.44	−40.6%	22.07	+15.1%	25.0	+19.6%	27.17
Feb	9.01	−44.7%	19.95	+11.8%	24.0	+28.3%	25.38
Mar	9.61	−43.8%	18.79	−3.8%	26.8	+13.1%	26.05
Apr	9.54	−47.4%	17.28	−7.2%	26.2	+2.3%	25.05
May	9.61	−45.7%	18.40	−11.3%	26.1	+4.3%	25.69
Jun	10.10	−33.1%	20.93	+1.6%	25.5	−1.9%	27.40

Implied demand = 1.08× polysilicon + 0.52× DMC + 0.22× primary alloy, excluding exports.

Polysilicon output fell by 33–47% year on year as the solar industry cut back after overbuilding in 2024. DMC output was flat to slightly down. Primary alloy output grew strongly early in 2025 (+20% to +28%) before flattening by June.

Chapter 4

Supply–Demand Balance

4.1 Method

A monthly balance compares supply with demand estimated from downstream output, all in 10,000 t:

$$\text{Supply}_t = \text{MG-Si output}_t + 97\#\ \text{output}_t + \text{recycled Si}_t + \text{imports}_t \tag{4.1}$$

$$\text{Demand}_t = 1.08\,\text{Polysilicon}_t + 0.52\,\text{DMC}_t + 0.22\,\text{Primary alloy}_t + \text{exports}_t \tag{4.2}$$

$$\text{Surplus}_t = \text{Supply}_t - \text{Demand}_t \tag{4.3}$$

4.2 Results

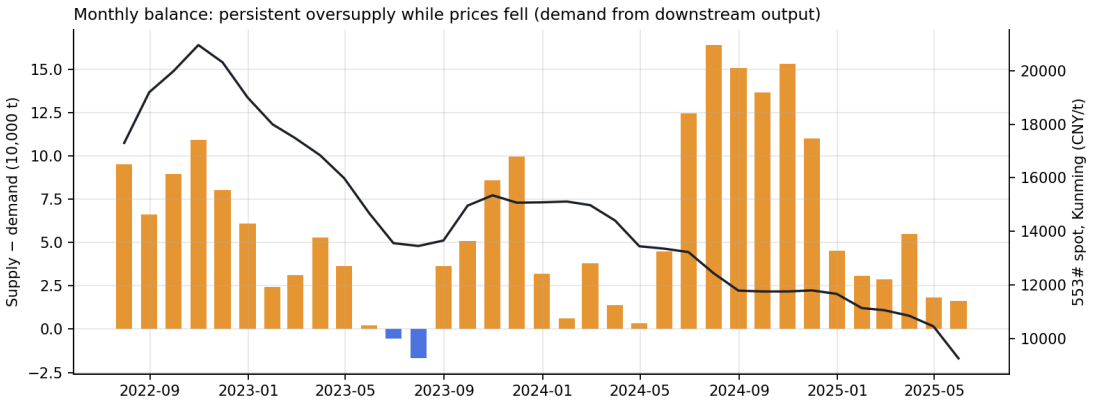


Figure 4.1: Monthly surplus of supply over estimated demand (bars) and 553# spot price in Kunming (line), July 2022 – May 2025.

Estimated supply exceeded demand in **33 of 35 months**, by an average of about 59,000 t a month. The largest surpluses came in the second half of 2024, when Xinjiang output peaked while polysilicon producers were cutting back. The persistent surplus is consistent with the fall in spot prices from about 21,000 CNY/t in 2022 to under 10,000 CNY/t in 2025.

A caveat: the balance does not reconcile with inventory

If the estimated surplus were right, inventories would have grown by about 59,000 t a month. Reported social inventory (stocks held in third-party warehouses) rose from 99,000 t in July 2022 to 551,000 t in July 2025, about 12,600 t a month. Either some demand is not captured by the three downstream sectors (for example, smaller uses or under-reported output), or much of the surplus is held at plants and by traders rather than in social warehouses. The balance is therefore more reliable for its *direction* than for its *level*.

Chapter 5

Production Costs

5.1 The cost model

Total cost per tonne is the sum of input costs, each equal to unit consumption times unit price:

$$C_{\text{total}} = \sum_{i} q_i \, p_i = C_{\text{silica}} + C_{\text{carbon}} + C_{\text{electrode}} + C_{\text{electricity}} + C_{\text{labour}} + C_{\text{other}} \tag{5.1}$$

where q_i is consumption per tonne of silicon (tonnes or kWh) and p_i the unit price. C_{other} covers maintenance, logistics, environmental fees, taxes and depreciation. Consumption figures come from plant feasibility studies, environmental impact assessments and project disclosures; prices are triangulated from SMM, broker research and industry publications. All figures include tax.

5.2 Regional and company cost structures

Table 5.1: Production cost by region and for the lowest-cost producer (CNY/t)

Cost item	Hoshine, Shanshan (2024)	Xinjiang average	Southwest wet season	Southwest dry season
Silica	1,080	1,080	1,375	1,375
Coal / coke	1,485	1,485	1,260	1,260
Petroleum coke	–	–	1,040	1,040
Wood chips	200	300	225	225
Graphite electrode	855	950	960	960
Electricity	**3,080**	**3,680**	**4,200**	**6,360**
kWh/t × CNY/kWh	11,000 × 0.28	11,500 × 0.32	12,000 × 0.35	12,000 × 0.53
Labour	400	400	400	400
Other expenses	1,000	1,000	800	800
Environmental fees	200	200	200	200
Sales, admin, R&D	–	150	–	–
Total	**8,300**	**9,245**	**10,460**	**12,620**
Electricity share	37%	40%	40%	50%

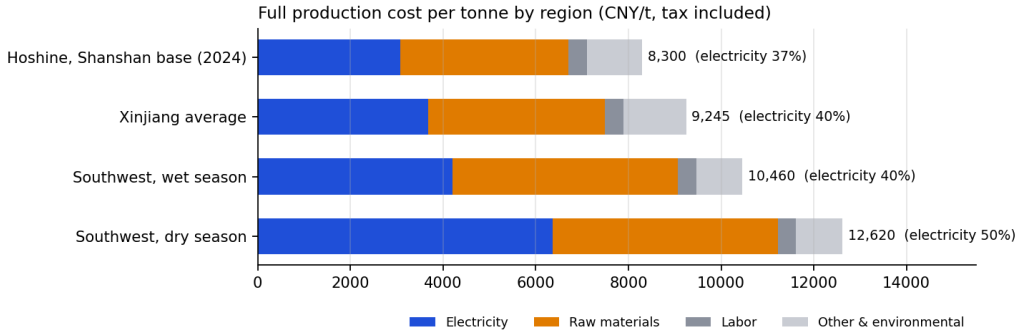


Figure 5.1: Composition of full production cost by region.

- **Xinjiang** producers typically spend 9,000–10,000 CNY/t. Cheap coal power offsets higher logistics costs to eastern customers.
- **Southwest** producers spend 9,500–13,000 CNY/t, depending on the season. Electricity alone moves cost by about 2,200 CNY/t between the wet and dry seasons.
- **Hoshine**, the largest producer, keeps costs near 8,300 CNY/t through scale, integration and its own power plants.
- **Secondary regions** (Fujian, Guizhou, Hunan) lack these cost advantages but survive by serving export markets (Fujian's ports) or local customers, and by producing higher grades with tight impurity limits.

5.3 Controlling electricity costs

Captive power plants. Grid power for industry costs 0.30–0.40 CNY/kWh, while producers' own coal-fired plants supply power at 0.18–0.26 CNY/kWh. At 11,500–12,000 kWh/t, this saves 700–1,400 CNY/t. Captive plants face growing emissions limits and are not permitted in every province, but remain the dominant cost advantage in Xinjiang and Inner Mongolia.

Waste heat recovery. Furnace off-gases can raise steam to drive a turbine (Figure 5.2). Environmental impact assessments of plants in Inner Mongolia, Sichuan and Yunnan report that recovered heat supplies about **20%** of a plant's power. One plant with a 2 × 20 MW system generated 297.8 million kWh of its 1.24 billion kWh consumption (24%). High capital cost and the need for stable furnace loads keep adoption below its potential.

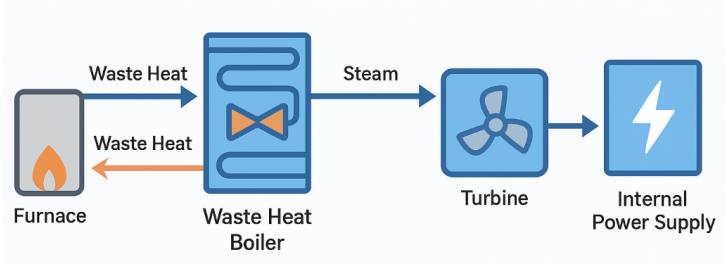


Figure 5.2: Waste heat recovery: furnace off-gas heat raises steam, which drives a turbine to supply the plant's own power.

5.4 Electricity sensitivity

Varying the tariff from 0.25 to 0.35 CNY/kWh and consumption from 10,000 to 13,000 kWh/t around Hoshine's cost structure gives total costs from 7,720 to 9,770 CNY/t (Figure 5.3).

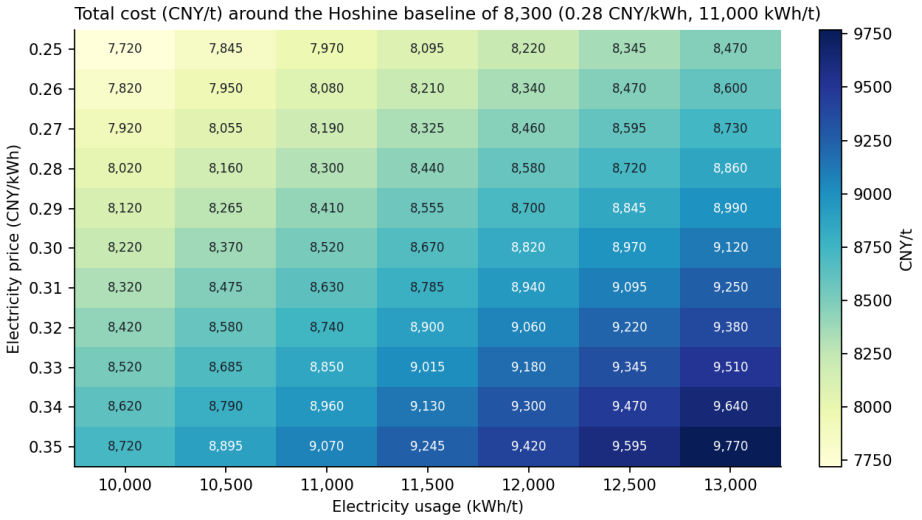


Figure 5.3: Total cost (CNY/t) by electricity tariff and consumption, around Hoshine's baseline.

Elasticity of cost to the electricity price equals electricity's cost share

The elasticity ε measures the percentage change in total cost for a 1% change in the power price. With consumption q fixed, $C = qp + C_{\text{rest}}$, so

$$\varepsilon = \frac{\partial C}{\partial p} \cdot \frac{p}{C} = \frac{q\,p}{C} = \text{electricity's share of total cost.}$$

At Hoshine's baseline, $\varepsilon = 3{,}080/8{,}300 = 0.371$: a 1% rise in the power price raises total cost by 0.37%. The same rule gives 0.40 for an average Xinjiang plant and 0.50 for Southwest plants in the dry season.

Chapter 6

Cost Curve, Margins and Stress Tests

6.1 The industry cost curve

Sorting producers from lowest to highest cost and stacking their capacity gives the industry's cost curve, or supply curve. When the price falls below a producer's cost, that producer loses money on every tonne and, over time, cuts output.

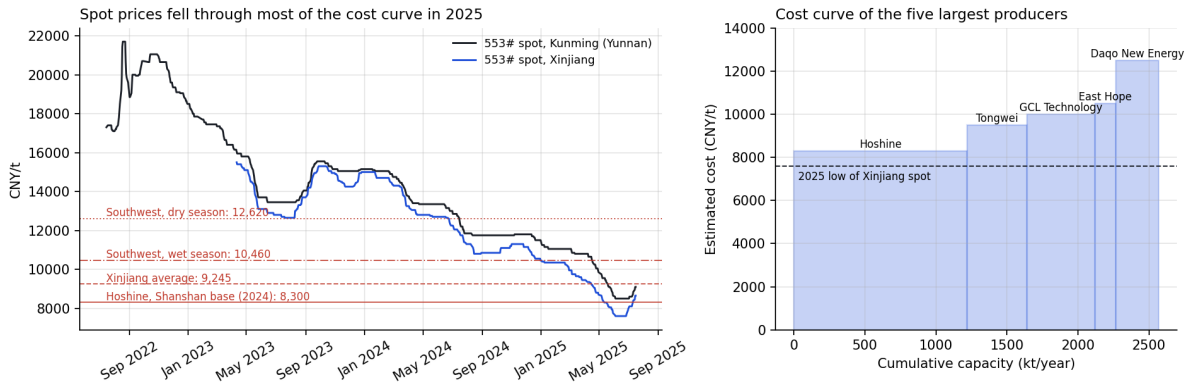


Figure 6.1: Left: 553# spot prices against the cost of each producer type. Right: cost curve of the five largest producers, with the 2025 low in Xinjiang spot.

Spot prices fell through almost the whole cost curve. On 4 June 2025 Xinjiang 553# reached **7,600 CNY/t**, below the 8,300 CNY/t cost of Hoshine, the lowest-cost producer. With nearly the whole industry losing money, prices rallied in July 2025 after government statements against "involution" (destructive price competition), not because measured supply or demand had changed.

6.2 Margins at reference prices

Margin is the selling price minus full production cost. Xinjiang producers are valued at Xinjiang 553# spot and Southwest producers at Kunming 553# spot.

Table 6.1: Margins at three reference prices (CNY/t; percentage of price in brackets)

Producer	Full cost	2024 average	2025 low (4 Jun 2025)	Latest (16 Jul 2025)
Selling price, Xinjiang / Kunming		12,348 / 12,936	7,600 / 8,500	8,650 / 9,100
Hoshine, Shanshan	8,300	4,048 (33%)	−700 (−9%)	350 (4%)
Xinjiang average	9,245	3,103 (25%)	−1,645 (−22%)	−595 (−7%)
Southwest, wet season	10,460	2,476 (19%)	−1,960 (−23%)	−1,360 (−15%)
Southwest, dry season	12,620	316 (2%)	−4,120 (−48%)	−3,520 (−39%)

Margins collapsed from 19–33% in 2024 to losses for all producer types in June 2025. Even after the July rally, only the lowest-cost producer was profitable.

A useful summary is the **breakeven electricity tariff**, the power price at which margin is zero given the selling price and all other costs:

$$p^{*}_{\text{elec}} = \frac{P - C_{\text{non-electricity}}}{q_{\text{elec}}} \tag{6.1}$$

At July 2025 prices it is 0.31 CNY/kWh for Hoshine (which pays 0.28), 0.27 for an average Xinjiang plant (which pays 0.32), and 0.24 for Southwest plants, below even their wet-season tariff of 0.35.

6.3 Stress tests

Each driver is shocked by a plausible amount, alone and in combination, starting from July 2025 prices:

- **Electricity tariff:** ±0.05 CNY/kWh, roughly the gap between captive and grid power, or between regions.
- **Raw materials** (silica, carbon reductants, electrodes): ±20%.
- **Product price:** ±10%, about one month's typical move (Chapter 8).

Table 6.2: Margin under shocks, starting from July 2025 prices (CNY/t)

Scenario	Hoshine Shanshan	Xinjiang average	Southwest wet	Southwest dry
Base	350	−595	−1,360	−3,520
Electricity +0.05 CNY/kWh	−200	−1,170	−1,960	−4,120
Electricity −0.05 CNY/kWh	900	−20	−760	−2,920
Raw materials +20%	−374	−1,358	−2,332	−4,492
Raw materials −20%	1,074	168	−388	−2,548
Product price −10%	−515	−1,460	−2,270	−4,430
Product price +10%	1,215	270	−450	−2,610
All three adverse	−1,789	−2,798	−3,842	−6,002
All three favourable	2,489	1,608	1,122	−1,038

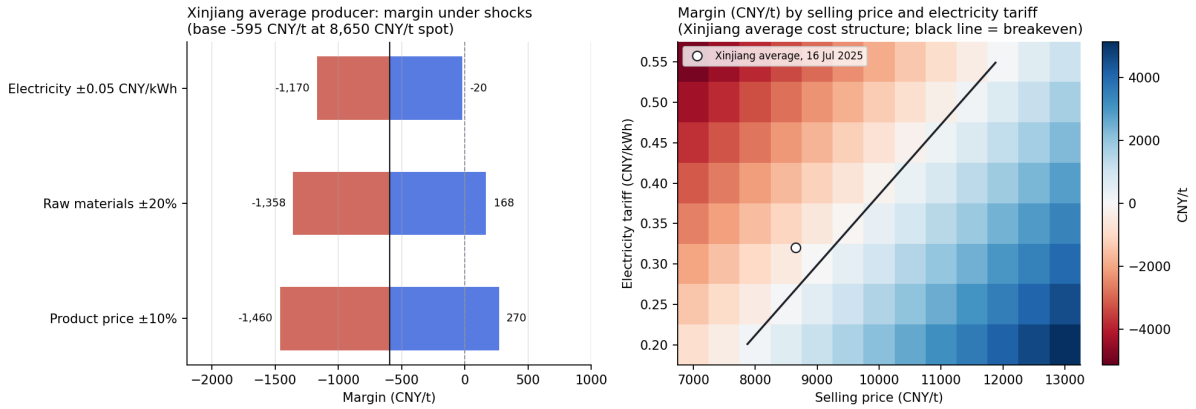


Figure 6.2: Left: margin of an average Xinjiang producer under each shock. Right: margin across selling prices and electricity tariffs; the black line marks breakeven.

In brief

- The product price is the largest single risk: a 10% move shifts margin by 865–910 CNY/t. Electricity (±550–600 CNY/t) and raw materials (±725–970 CNY/t) are of similar size.
- At July 2025 prices, *any one* adverse shock turns Hoshine's thin profit into a loss.
- Southwest plants in the dry season lose money even if all three drivers move in their favour, which explains why they cut output when hydropower is scarce.
- Each 0.05 CNY/kWh of tariff moves Xinjiang's breakeven selling price by about 575 CNY/t(the slope of the breakeven line in Figure 6.2).

Chapter 7

Price Forecasting with ARIMA

7.1 Data, and a hidden contract switch

The data are daily closing prices, trading volume and open interest for GFEX industrial silicon futures from 22 December 2022 to 21 July 2025, as a continuous "active contract" series from Wind. Futures contracts expire, so a continuous series is stitched together from different delivery months, switching as trading moves from one contract to the next.

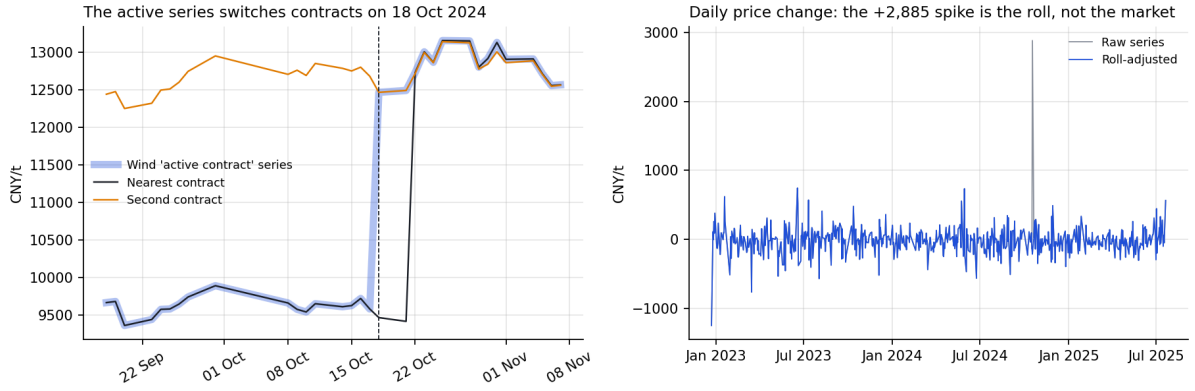


Figure 7.1: Left: on 18 October 2024 the active series switches from the nearest contract to the next, priced about 3,100 CNY/t higher. Right: the resulting spike in daily price changes, before and after adjustment.

Most switches cross small price differences of 0–3%. On **18 October 2024**, the switch crossed an unusually large one. That day the nearest contract fell from 9,580 to 9,465 and the next contract from 12,680 to 12,465, but the active series jumped from 9,580 to 12,465, a **false +30.1% move**. No other day moved by more than 8.3%.

The fix is to *back-adjust* prices before the switch by the gap between the two contracts. For models of price levels, earlier prices are shifted by +3,100 CNY/t (difference adjustment); for models of percentage returns, they are scaled by 12,680/9,580 (ratio adjustment). This one correction changes the statistical picture substantially:

Table 7.1: Daily log returns before and after correcting the contract switch

	Raw series	Adjusted
Kurtosis (normal distribution = 3)	75.4	5.8
Skewness	5.04	0.17
ARCH-LM test for volatility clustering, p-value	1.00	0.004
Variance of daily price changes, training period (CNY/t)^2	51,986	35,357

7.2 How ARIMA works

ARIMA in plain terms

An **ARIMA(p, d, q)** model describes a series using three ingredients:

- **I (integrated), order d:** difference the series d times until it has no trend. Prices wander without a fixed level, so we use $d = 1$ and model the daily change $w_t = y_t - y_{t-1}$.
- **AR (autoregressive), order p:** today's change depends on the previous p changes.
- **MA (moving average), order q:** today's change depends on the previous q forecast errors ("surprises").

In full, with a constant c (drift) and errors ε_t:

$$w_t = c + \phi_1 w_{t-1} + \cdots + \phi_p w_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}.$$

So **an ARIMA$(p$,1,$q)$ model of prices is an ARMA(p, q) model of price changes**; it is already a model of returns. Fitting it to log prices makes w_t the daily log return.
The **random walk** is the special case ARIMA(0,1,0): price changes are unpredictable noise, so the best forecast of tomorrow's price is today's. Every other ARIMA model is a claim that some of tomorrow's change *is* predictable from past changes.

Why forecasting prices and forecasting returns are the same test

Tomorrow's price is today's known price plus tomorrow's change: $\hat{y}_{t+1} = y_t + \hat{w}_{t+1}$. The price forecast error is therefore exactly the return forecast error, $y_{t+1} - \hat{y}_{t+1} = w_{t+1} - \hat{w}_{t+1}$. A model "beats the random walk" on prices if and only if its return forecast $\hat{w}_{t+1}$ beats a forecast of zero. Price forecasts always *look* accurate (the error is small relative to the price level) because most of the forecast is today's price; the real question is whether the small predicted change adds anything.

7.3 Stationarity and model identification

The Augmented Dickey–Fuller (ADF) test checks whether a series has a unit root, i.e. wanders without returning to a fixed level. Prices do ($p = 0.86$), while daily changes do not ($p < 0.001$), so one order of differencing is enough ($d = 1$).

The autocorrelation function (ACF) measures how strongly changes are correlated with changes k days earlier; the partial autocorrelation function (PACF) does the same after removing the effect of the intermediate days. Neither shows any meaningful pattern (Figure 7.2): the first two log-return autocorrelations are -0.07 and -0.08, inside the 95% band of ± 0.088.

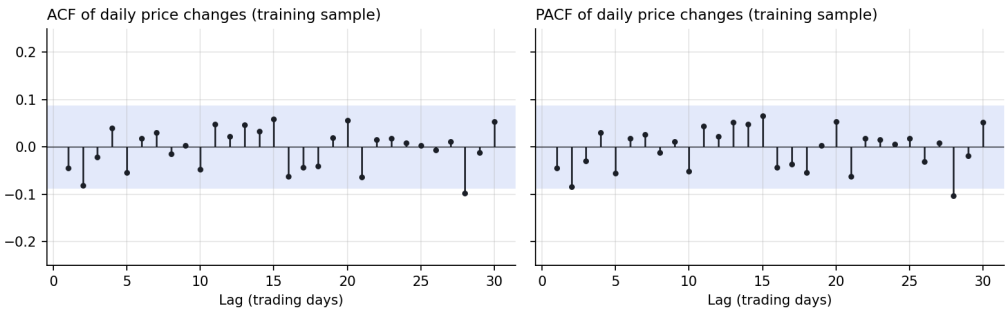


Figure 7.2: Autocorrelation and partial autocorrelation of daily price changes (training period), with 95% bands.

An exhaustive search over AR and MA orders up to 2, with and without drift, ranks models by the Akaike Information Criterion (AIC), which rewards fit and penalises extra parameters. All candidates lie within 2 AIC points of each other, including a plain random walk with drift (0.7 points from the best), which means the data cannot distinguish them. The original ARIMA(0,1,1) gives an MA coefficient of $\theta_1 = -0.037$ (standard error 0.048, $p = 0.44$): statistically indistinguishable from zero.

7.4 Out-of-sample evaluation

The sample is split in time: models are estimated on 22 December 2022 – 10 January 2025 (498 trading days) and evaluated on 13 January – 21 July 2025 (125 days). Two forecasting designs are used:

- **Static multi-step:** fit once and forecast the whole test period, as a trader would if never updating the model.
- **Rolling one-step:** each day, re-estimate on all data up to yesterday and forecast today's close.

The Diebold–Mariano test

Comparing two forecasts' average errors is not enough: with 125 days, one model can win by luck. The Diebold–Mariano (DM) test asks whether the difference in average squared error between two forecasts is larger than its day-to-day noise would produce by chance. A p-value below 0.05 indicates a statistically reliable difference.

Table 7.2: Rolling one-step forecasts over the test period (13 Jan – 21 Jul 2025)

Model	RMSE (CNY/t)	vs random walk	DM p-value	Direction correct
Random walk (tomorrow = today)	144.9	–	–	–
ARIMA(0,1,1) on prices	143.9	−0.7%	0.10	52%
ARIMA(2,1,0) on prices	145.6	+0.5%	0.35	48%
ARMA(0,1) on log returns	–	OOS $R^2 = +1.6\%$	0.23	52%
ARMA(1,0) on log returns	–	OOS $R^2 = +1.7\%$	0.18	53%
Lagged open interest & volume	–	OOS $R^2 = -0.6\%$	–	–

RMSE: root mean squared error. OOS R^2: share of the variation in daily returns explained out of sample, relative to forecasting zero (the random walk).

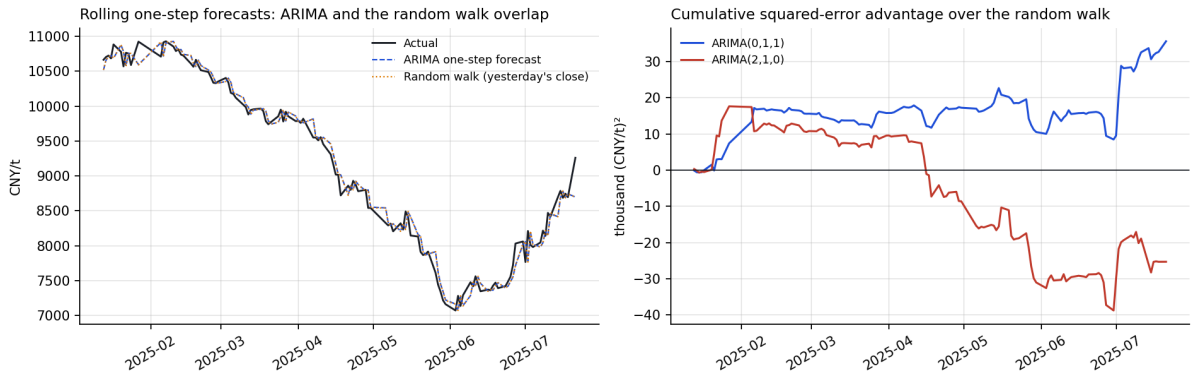


Figure 7.3: Left: rolling one-step forecasts overlap the random walk. Right: cumulative squared-error advantage over the random walk; ARIMA(0,1,1)'s small lead comes from a few days in February and July.

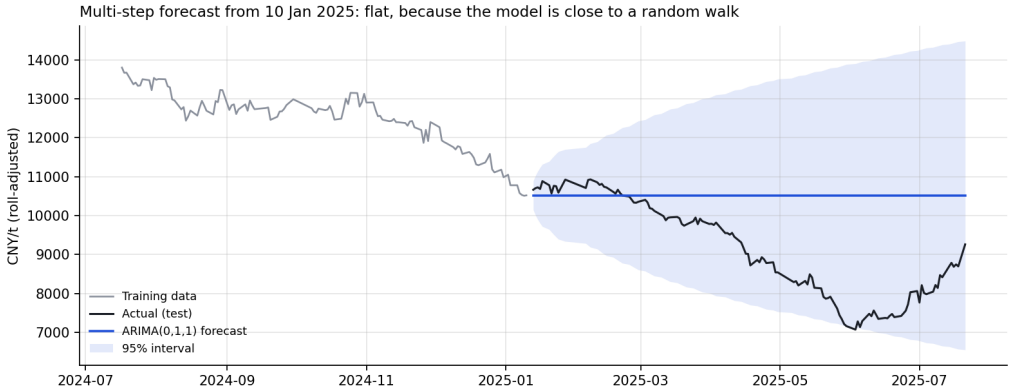


Figure 7.4: Static multi-step forecast from 10 January 2025. The forecast is flat because the model is essentially a random walk; RMSE 1,867 CNY/t, MAE 1,498 CNY/t.

7.5 Interpretation

- **Rolling forecasts track prices closely, but so does the random walk.** The tracking comes from using yesterday's close, not from the model. No specification beats the random walk by a statistically significant margin.
- **Returns show a hint of short-term reversal.** Slightly negative autocorrelation at lags 1–2 lets ARMA models on log returns explain 1.6–1.7% of next-day return variation out of sample. In equity markets an R^2 of this size can be economically meaningful, but here it is not statistically significant over 125 days and needs a longer sample to confirm.
- **Static forecasts cannot anticipate trends.** A near-random-walk model's best multi-step forecast is a flat line at the last price, so it missed both the decline to June and the July rally.
- **Price-level intervals mislead.** Rolling 95% intervals on prices contained 98.4% of outcomes, apparently conservative. But they measure risk in CNY/t, and prices roughly halved between the training and test periods; in percentage terms volatility *rose*, from 1.42% to 1.68% a day. Risk is better modelled on returns, as in the next chapter.

Chapter 8

Volatility Forecasting with GARCH

8.1 Why model volatility?

Even when the direction of tomorrow's move is unpredictable, its *size* may not be: calm and turbulent periods tend to persist. This is called **volatility clustering**. It matters for risk limits, margin requirements, hedge sizing and the stress tests of Chapter 6. The ARCH-LM test (Table 7.1) confirms clustering in the corrected data ($p = 0.004$).

8.2 The models

All models describe the daily log return r_t (in %) as a constant mean plus a shock, $r_t = \mu + \varepsilon_t$, with $\varepsilon_t = \sigma_t z_t$, where σ_t^2 is the **conditional variance**: the variance expected for day t given everything known on day $t-1$. They differ in how σ_t^2 is updated.

Constant variance

$\sigma_t^2 = \sigma^2$. Every day is assumed equally risky. This is the assumption behind ordinary ARIMA forecast intervals.

EWMA (exponentially weighted moving average, "RiskMetrics")

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,\varepsilon_{t-1}^2, \qquad \lambda = 0.94.$$

Tomorrow's variance is a weighted average of today's variance estimate and today's squared shock. With $\lambda = 0.94$, a shock's weight halves after about 11 days. Developed by J.P. Morgan for its RiskMetrics system, EWMA needs no estimation, but it has no long-run average: volatility drifts wherever recent shocks push it and never mean-reverts.

GARCH(1,1) (generalised autoregressive conditional heteroskedasticity)

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2.$$

Like EWMA, but with a constant ω that pulls volatility back towards a long-run level, $\omega/(1-\alpha-\beta)$.

- α (reaction): how strongly yesterday's surprise raises today's variance.
- β (memory): how much of yesterday's variance carries over.
- $\alpha+\beta$ (**persistence**): how slowly shocks fade. The **half-life**, $\ln 0.5/\ln(\alpha+\beta)$, is the number of days for a volatility shock to halve.

EWMA is the special case $\omega = 0$, $\alpha+\beta = 1$.

GJR-GARCH (Glosten–Jagannathan–Runkle)

$$\sigma_t^2 = \omega + \big(\alpha + \gamma\,\mathbf{1}[\varepsilon_{t-1} < 0]\big)\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2.$$

Adds an asymmetry term: $\mathbf{1}[\cdot]$ equals 1 on days with a negative shock, so falls raise volatility by $\alpha+\gamma$ and rises by α. In equities $\gamma > 0$ (the "leverage effect": falls raise volatility more). In commodities, supply shocks and short squeezes can make $\gamma < 0$ instead, with rallies the more volatile.

Normal versus Student-t errors

The standardised shock z_t can follow a normal distribution or a Student-t distribution with ν degrees of freedom. The t distribution has fatter tails: extreme days are more frequent, increasingly so as ν falls. ν below about 5 indicates very fat tails; as ν grows the t approaches the normal.

8.3 In-sample fit

Models are estimated by maximum likelihood on the 497 daily returns of the training period.

Table 8.1: Volatility models, training period. ΔAIC is the distance from the best model (lower is better). The last column tests whether clustering remains in the standardised residuals ε_t/σ_t; a high p-value means the model has captured it.

Model	Log-lik.	ΔAIC	Parameters	Clustering left (p)
Constant variance, normal	−877.5	72.3		0.008
Constant variance, t	−845.1	9.5	$\nu = 3.8$	0.008
EWMA, $\lambda = 0.94$	−883.7	82.6		0.95
GARCH(1,1), normal	−871.3	63.8	$\alpha = 0.03$, $\beta = 0.93$	0.41
GARCH(1,1), t	**−838.4**	**0.0**	$\alpha = 0.05$, $\beta = 0.93$, $\nu = 3.5$	0.72
GJR-GARCH(1,1), t	−838.2	1.6	$\gamma = -0.08$, $\nu = 3.6$	0.71

- **Fat tails matter most.** Moving from normal to Student-t errors improves AIC by about 60 points. With $\nu \approx 3.5$, extreme days are far more frequent than a normal distribution allows.
- **Clustering matters next.** GARCH-t improves on constant-variance-t by 9.5 AIC points and removes the remaining clustering. Persistence $\alpha+\beta = 0.98$ implies a **half-life of**

about 35 trading days: after a turbulent episode, volatility takes roughly seven weeks to fall halfway back to normal.

- **No reliable asymmetry.** GJR estimates $\gamma < 0$, so rallies would raise volatility more than falls, plausible for a policy- and supply-driven commodity. But the extra parameter barely improves the fit, and AIC prefers the symmetric model.

8.4 Out-of-sample evaluation

Each model is re-estimated daily over the test period and produces a one-day-ahead variance forecast and a 95% interval for the next day's return.

Judging volatility forecasts: QLIKE and backtests

- **QLIKE loss**, $\ln \sigma_t^2 + r_t^2/\sigma_t^2$, scores a variance forecast σ_t^2 against the realised squared return. Lower is better. It is the standard loss for volatility because it stays reliable even though a single day's squared return is a very noisy measure of that day's true variance.
- **Breaches**: a 95% interval should be missed on about 5% of days, or 6.25 of 125.
- **Kupiec test**: is the number of breaches consistent with 5%? A low p-value rejects the model.
- **Christoffersen independence test**: do breaches cluster on consecutive days? A low p-value means the model reacts too slowly.

Table 8.2: One-day-ahead volatility forecasts, 13 January – 21 July 2025

Model	QLIKE	DM p vs constant	Breaches (expect 6.3)	Kupiec p	Interval width
Constant variance	2.089	–	13	0.015	5.5%
EWMA	1.926	0.17	11	0.08	5.5%
GARCH(1,1), normal	1.943	0.06	13	0.015	5.6%
GARCH(1,1), t	**1.920**	**0.08**	**10**	**0.16**	**6.0%**
GJR-GARCH(1,1), t	1.958	0.11	11	0.08	6.0%

Christoffersen independence p-values are 0.74–0.90 for all models: breaches do not cluster.

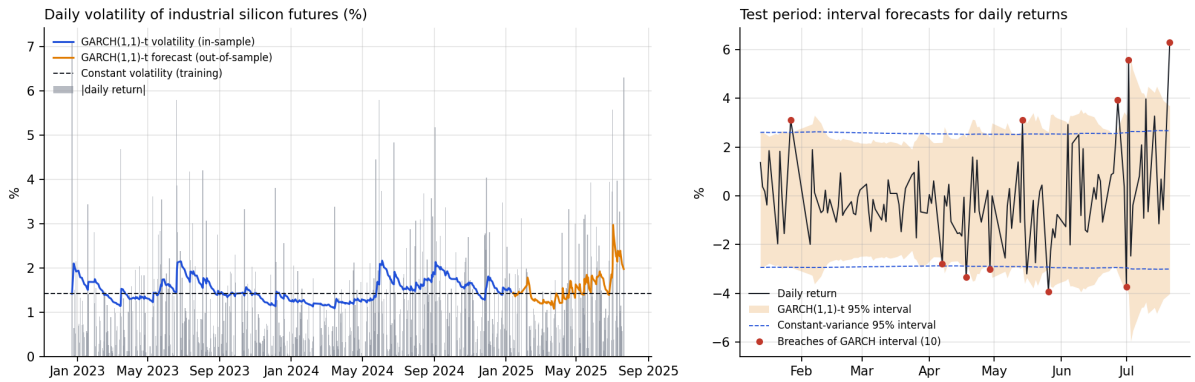


Figure 8.1: Left: GARCH(1,1)-t volatility in the training period and forecasts in the test period, with absolute daily returns. Right: 95% intervals for daily returns in the test period and the days that breached the GARCH-t interval.

8.5 Interpretation

- **GARCH(1,1)-t is the best model.** It has the lowest QLIKE loss and is the only model whose 95% intervals pass the Kupiec test. Constant variance and normal GARCH breach twice as often as they should and are rejected.
- **The evidence is modest.** QLIKE improvements over constant variance are 6–8%, with DM p-values of 0.06–0.17 over 125 days.
- **Policy shocks defeat every model.** Four of GARCH-t's ten breaches fall between 27 June and 21 July 2025, around the anti-involution rally, including the two largest moves of the test period (+5.6% and +6.3%). GARCH raises its forecast only *after* large moves, reaching about 3% a day by July, so it cannot anticipate policy announcements.

8.6 Application: how likely are producers to lose money next month?

The GARCH model links the statistical and fundamental analyses. Re-estimated on the full sample, GARCH(1,1)-t forecasts a next-day volatility of 2.7% as of 21 July 2025, well above average after the rally. Simulating 20,000 paths of 21 trading days (one month) from this starting point gives a 90% range for the monthly price change of −18% **to** +18%, against ±11% under constant volatility with normal errors. Applying these paths to July 2025 spot prices gives the distribution of each producer's margin one month ahead.

Table 8.3: Margin one month ahead, from 20,000 simulated GARCH(1,1)-t price paths (CNY/t)

Producer	P(loss)	5th percentile	Median	95th percentile
Hoshine, Shanshan	33%	−1,095	351	2,059
Xinjiang average	75%	−2,040	−594	1,114
Southwest, wet season	91%	−2,880	−1,359	438
Southwest, dry season	99%	−5,040	−3,519	−1,722

Assumes no drift in prices, futures volatility applied to spot prices, and costs fixed at the levels in Table 5.1.

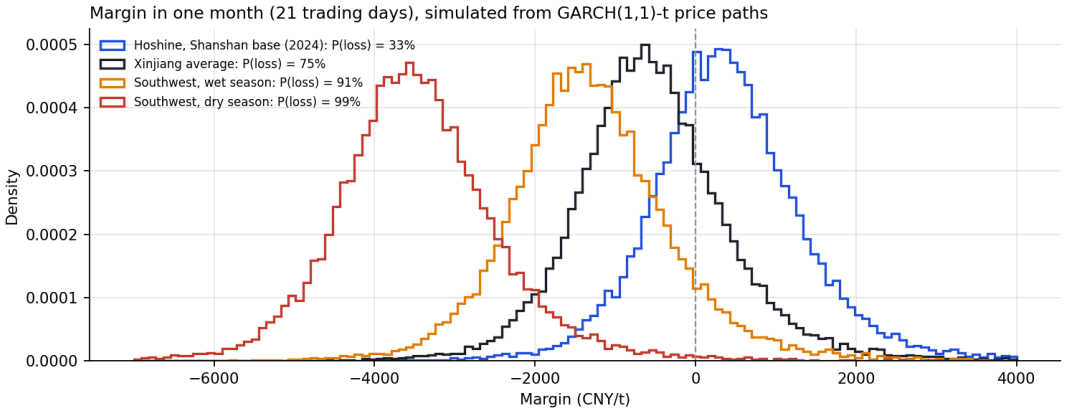


Figure 8.2: Distribution of margin one month ahead for each producer type.

Using constant volatility would understate the spread of outcomes by about 40%, and with

it the risk of losses for the lowest-cost producer.

Chapter 9

Conclusions

9.1 Summary of models

Table 9.1: What each model found

Model	Result	Interpretation
Random walk	RMSE 144.9 CNY/t (1.2% of price)	The benchmark every price forecast must beat
ARIMA on prices	−0.7% RMSE, not significant	Daily price changes have almost no memory
ARMA on returns	OOS R^2 1.6–1.7%, not significant	A possible slight short-term reversal, too weak to confirm
Open interest & volume	OOS $R^2 = -0.6\%$	Positioning carries no next-day signal
Constant variance	13 breaches of 95% interval; rejected	Understates risk in turbulent periods
EWMA	11 breaches; borderline	Adapts, but never reverts to a normal level
GARCH(1,1)-t	**Best fit and forecasts; passes backtest**	**Volatility clusters (half-life ≈ 35 days); tails are fat ($\nu \approx 3.5$)**
GJR-GARCH-t	No gain over GARCH	No robust asymmetry between rises and falls
Cost model & stress tests	Prices below most costs in mid-2025	Thin margins make producers highly sensitive to every driver

9.2 Implications

The results fit an efficient but policy-sensitive market:

- **Direction is not forecastable from past prices**, but **risk is**. For hedgers and risk managers, GARCH-t volatility forecasts are the useful output, not price forecasts.
- **Fundamentals show where prices are unsustainable**, but not when they will turn. With prices below the costs of nearly every producer by June 2025, either output cuts or policy intervention had to follow; it came through policy in July.
- **Electricity is the decisive competitive factor.** Producers with captive power or wet-season hydropower survive downturns that force others to shut.

9.3 Limitations

- Costs are estimates triangulated from public documents and market data, not audited company figures, and use one year's input prices.
- The test period (125 days) is short, so modest forecasting differences cannot be confirmed statistically.
- The supply–demand balance does not reconcile with reported inventory.
- The margin simulation applies futures volatility to spot prices and holds costs fixed.

9.4 Future work

- **A policy and sentiment index.** Prices in July 2025 rose sharply on government meetings and statements against involution, while capacity and demand were unchanged. Text mining of official communications, combined with event studies, could measure the tone and timing of policy signals.
- **Volatility around policy events.** Jump components, or indicators for scheduled policy meetings, could address the breaches GARCH still shows in July 2025; intraday data would allow realised-volatility models.
- **Reconciling the balance sheet** with inventory data, and testing whether the surplus or the price's distance from the cost curve predicts monthly returns.
- **The term structure.** The 3,100 CNY/t gap between adjacent contracts in October 2024 suggests the futures curve contains information about expected supply.
- **Nonlinear models** such as gradient boosting or recurrent neural networks on fundamental and positioning data, held to the same random-walk benchmark.

Appendix A

Data Sources and Glossary

A.1 Data

Data	Source	Frequency and period
Futures, active contract: close, volume, open interest	Wind	Daily, 22 Dec 2022 – 1 Aug 2025
Nearest and second-nearest contracts	Wind	Daily
553# spot prices (Xinjiang, Kunming, Qinghai, Inner Mongolia)	SMM	Daily, from July 2022
Output, operating rates, downstream production	SMM	Monthly and weekly, from July 2022
Exports, imports, social inventory	SMM	Monthly and weekly
Unit consumption of inputs	Feasibility studies, environmental impact assessments	–
Input prices	SMM, broker research, industry publications	2024–2025

The code for every figure and table is available at

`https://github.com/dengyuhe/industrial-silicon-futures`

The vendor data are licensed and not redistributed.

A.2 Glossary of market terms

Chinese	English
工业硅 / 金属硅	Industrial silicon / metallurgical silicon
多晶硅	Polysilicon
有机硅	Organosilicon (silicones)
原生铝合金 / 再生铝合金	Primary / recycled aluminium alloy
活跃合约	Active (most-traded) contract
开工率	Operating rate
社会库存 (社库)	Social inventory (third-party warehouses)
供需平衡表	Supply–demand balance sheet
丰水期 / 枯水期	Wet / dry season (hydropower)
自备电厂	Captive power plant
余热发电	Waste heat power generation
反内卷	Anti-involution (against destructive price competition)